

	Answer	Mark
6(a)(i)	The only correct answer is B. <i>A is incorrect because P is a ribosome</i> <i>C is incorrect because R is a thylakoid</i> <i>D is incorrect because S is a starch grain</i>	(1)

	Answer	Mark
6(a)(ii)	The only correct answer is C. <i>A is incorrect because starch is stored in S</i> <i>B is incorrect because starch is stored in S</i> <i>D is incorrect because starch is stored in S</i>	(1)

	Answer	Mark
6(a)(iii)	The only correct answer is B. <i>A is incorrect because</i> <i>C is incorrect because</i> <i>D is incorrect because</i>	(1)

	Answer	Mark
6(a)(iv)	The only correct answer is A. B is incorrect because P is $0.02\mu\text{m}$, R is $0.435\mu\text{m}$ and S is 1 to $35\mu\text{m}$ C is incorrect because P is $0.02\mu\text{m}$, R is $0.435\mu\text{m}$ and S is 1 to $35\mu\text{m}$ D is incorrect because P is $0.02\mu\text{m}$, R is $0.435\mu\text{m}$ and S is 1 to $35\mu\text{m}$	(1)

	Answer	Mark
6(b)(i)	The only correct answer is C. A is incorrect because CO_2 is used B is incorrect because CO_2 is used D is incorrect because area is mm^{-2} and time is hr^{-1}	(1)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	An answer that includes the following points: Similarities: • both have a rise and fall (in rate of photosynthesis with an increase in temperature) (1) • both have the same rate of photosynthesis at 16.5°C (1) Differences: • <i>Spartina</i> has {an optimum temperature of / highest rate at} 35°C whereas <i>Leucopoa</i> has {an optimum temperature of / highest rate at} 23°C (1) • <i>Leucopoa</i> has a higher rate below 16.5°C and <i>Spartina</i> has a higher rate above 16.5°C (1)	DO NOT PIECE TOGETHER ACCEPT 16.4°C ACCEPT values between 34 and 36, 21 and 24 a stated value difference provided it can be obtained from these values ACCEPT converse	(3)

Question number	Answer	Additional guidance	Mark
6(b)(iii)	A description that includes the following points: • read off the rate of photosynthesis at two temperatures 10°C apart (1) • divide the rate for the higher temperature by that for the lower temperature (1)	ACCEPT two stated temperatures e.g. 10°C and 20°C values for two rates at appropriate temperatures quoted {calculate / get} rate ACCEPT from an equation if t and t+10 have been described	(2)

Question number	Answer	Additional guidance	Mark
6(b)(iv)	An explanation that includes the following points: • Wheatland (1) • because Wheatland has the higher temperatures (throughout the whole year) (1) • therefore enzymes will be more active (at these higher temperatures) (1)	ACCEPT from a description appropriate named enzyme e.g. RUBISCO	(3)